



Instructor: Date:

Learner: Regional Branch:

Program: Center:

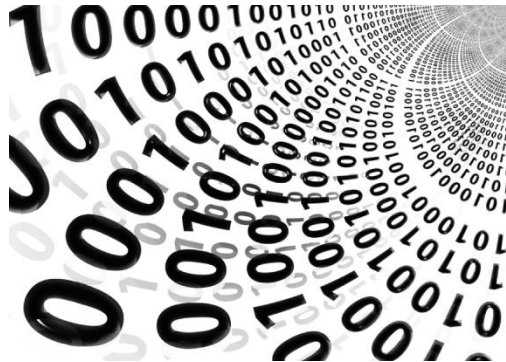
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Let's Start!

Mechanisms in Information Management Systems



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to query analysis, and data base engines. Describe relational database, its concepts, and principles in spoken or written texts.

Ask your partner:

- Do you think a database can allow several users to see and edit the same information at the same time? Why? Why not?
- How can we recover original files in a database after people have made several changes to them?

Can Do Statements:

- I can talk about the requirements all information management systems must have.
- I can discuss information about mechanisms like concurrency and read consistency.
- I can talk about mechanisms like catching and locking.



Let's Explore!

Sara and Michelle talk about concurrency.



Instructions: Read the text below.

Michelle: I wonder how Oracle databases can handle so much information at the same time.

Sara: Well, important requirements must be fulfilled. The data concurrency must be maximized; any data must be read and modified consistently because any improper change can be very dangerous, and maximum productivity must always be achieved.

Michelle: Are those databases supposed to be able to do all that? It sounds very difficult.

Sara: It is, but they have software mechanisms that help during the process. For example, they use concurrency, because many users have access to the same data at the same time. They will always be supposed to implement proper controls, otherwise, data could be updated or changed improperly, thus generating a serious impact on the system. They can also use read consistency.

Michelle: Wait a minute! I was supposed to learn about that last semester. What read consistency does is to make data available to be read or written, but without updating or changing the information for every single user, right?

Sara: Astonishing! That's exactly what it does. It creates private copies of the data; anyone can read or write it, but the data at the database remain unchanged until they are needed.

Michelle: I have heard that catching and locking are also supposed to be used.

Sara: Yeah, they use those, too. What catching allows is to create cache memory from user data, log data or even other types of data, and instead of repeating all processes, cache data is used whenever it is possible in order to save time.

Michelle: And what about locking?

Sara: The word itself explains it all! It locks the data since updates or deletes must be controlled to optimize productivity; locking allows you control the information that can be changed, it can be manual or automatic. It's up to you!





Let's Practice!

System Requirements

Instructions: Complete the sentences with the words from the box. Take turns with your partner and read the sentences aloud. Correct each other's pronunciation.

are supposed to / was supposed to allow / will be supposed to
modify / was supposed to improve / were supposed to be

1. Multiuser databases _____ handle different information at the same time.
2. Data concurrency _____ users to see and modify the same information at the same time.
3. In the future, stakeholders _____ consistently.
4. Last year, the programmer _____ consistency.
5. Transactions _____ executed at the same time when using concurrency mechanisms.



Let's Practice!

Consistency and Currency

Instructions: Match the sentences in column A with those in column B. Check your answers with your partner. Discuss.

COLUMN A

1. Data concurrency
2. Data consistency
3. Catching mechanism
4. Locking mechanism
5. Single-user database
6. Multiple-user database

COLUMN B

- a. This mechanism states that data must be edited and updated by several users.
- b. This mechanism states that data must be visible so that users can see changes in real time.
- c. This mechanism states that cache memory must be created by user data.
- d. This mechanism states that updates and deletes must be controlled to optimize productivity.
- e. Database that is supposed to be used by an individual user.
- f. Database that is supposed to allow multiple users to access data.



Extension Activity

Concurrency Features

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Write a list of mechanisms you are supposed to use to create a useful database.
- Describe the important features of the database you created.
- Explain the requirements that must be accomplished in order to manage information accurately.
- Present a report with the information above.

Self-Assessment

Yes

No

- I recognize mechanisms to manage information in databases.
- I identify the main features of single and multiple-user databases.
- I discuss about the requirements that must be accomplished in order to manage information accurately.

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Target Words

- **consistently** (adverb): to make something steadily without any change
e.g. The data concurrency must be maximized, any data must be read and modified consistently because any change is very dangerous.
- **concurrency** (noun): the simultaneous access of many users to the same data
e.g. They use concurrency, due to many users having access to the same data at the same time, they will always be supposed to implement proper controls.
- **astonishing** (adjective): surprising, striking
e.g. Astonishing! That's exactly what it does, it creates private copies of the data, anyone can read or write it.
- **lock** (verb): to make secure something in order to avoid people having access to it
e.g. The word itself explains it all! It locks the data, since updates or deletes must be controlled to optimize productivity.
- **be up to** (phrasal verb): to have the final decision about something
e.g. Locking lets you control the information that can be changed, it can be manual or automatic. It's up to you!



Learning Points

Be Supposed To in Present, Past, and Future

You can use **be supposed to** in any tense.

Present Tense

Use **be supposed to** in present to talk about actions that people are expecting from you, or that are always true

Structure

Subject + am / is / are + supposed to + verb + complement

Examples:

*I **am supposed to** wait for a turn to manage data concurrency.*

*Users **are supposed to** wait for a turn to manage data concurrency.*

Past Tense

Use **be supposed to** talk about actions that people expected in a certain condition in the past but are not longer true.

Structure

Subject + was / were + supposed to + verb + complement

Examples:

*I **was supposed to** wait for a turn to manage data concurrency.*

*Decades ago, users **were supposed to** wait for a turn to manage data concurrency.*

Future Tense

Use it to talk about actions people will expect from you when certain conditions are met.

Structure

Subject + will be + supposed to + verb + complement

Example:

*As a user, you **will be supposed to** wait for a turn to manage data concurrency.*

Examples

- *I question myself how oracle databases are supposed to handle so much information at the same time.*
- *Are they supposed to be able to do all that? That sounds very difficult.*
- *I was supposed to learn about that last semester*





Learning Points

Passive of Modal Verbs, Must

We use the **passive voice** when we want to emphasize on the action and not on the person who does it. **Passive voice** can be used anytime with **modal verbs**; we are going to focus on **must** this time.

Must and **Ought to** are similar.

They are used to talk about obligations, but **must** is more commonly used than **ought to**.

Structure

Subject + must be + verb in past participle + complement

Examples

That piece of evidence **must be used** in this case, it's of big importance

Any suggestion **must be written** in the suggestion box to be taken into account.

Examples

- *Important requirements must be followed. The data concurrency must be maximized, any data must be read and modified consistently because any change is very dangerous, and maximum productivity must always be achieved.*
- *The word itself explains it all! It locks the data, since updates or deletes must be controlled to optimize productivity, locking lets you control the information that can be changed.*

Data Concurrency and Consistency in a Multiuser Environment

Data concurrency and consistency are mechanisms that allow users to access a database and modify the data while other users are updating the same data at the same time.

Decades ago, databases were supposed to be single-user databases, which means that users were supposed to wait for a turn to manage data in an individual way.

Fortunately, databases have changed. Now, thanks to concurrency and consistency mechanisms, several users can update data and execute multiple simultaneous transactions in these multiuser databases.

Let's see a brief explanation of how data concurrency and data consistency work:

- Data **concurrency**: Allow many users to manage the same information at the same time.
- Data **consistency**: Allow users to see all changes in real time so that the database is always updated.

